Arquitecturas de Redes Neuronales

0) Bloques comunes

Base · Notación

Capa densa: $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{h} + \boldsymbol{b}, \quad \boldsymbol{h}' = \phi(\boldsymbol{z})$.
Activaciones: $\mathrm{ReLU}(x) = \text{máx}(0, x)$, $\sigma(x) = \frac{1}{1+e^{-x}}$, $\mathrm{softmax}(\boldsymbol{z})_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$.
Pérdidas típicas:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\boldsymbol{y}}_i - \boldsymbol{y}_i\|_2^2, \qquad \mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k} y_{ik}\log \hat{y}_{ik}.$$

1) MLP / Feedforward

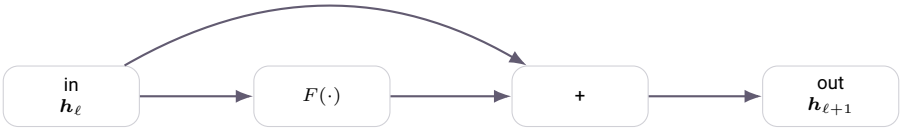
Denso

$$\boldsymbol{h}^{(\ell)} = \phi\Big(\boldsymbol{W}^{(\ell)}\boldsymbol{h}^{(\ell-1)} + \boldsymbol{b}^{(\ell)}\Big), \quad \hat{\boldsymbol{y}} = g(\boldsymbol{h}^{(L)}).$$

2) ResNet (conexión residual)

CNN · Residual

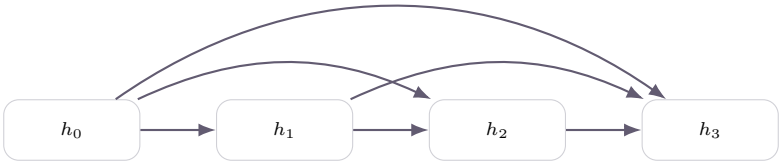
$$\boldsymbol{h}_{\ell+1} = \boldsymbol{h}_\ell + F(\boldsymbol{h}_\ell), \qquad \text{(skip connection)}$$



3) DenseNet (conexiones densas)

CNN · Concat

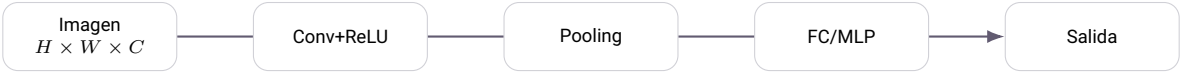
$$\boldsymbol{h}_\ell = H_\ell\big([\boldsymbol{h}_0, \boldsymbol{h}_1, \ldots, \boldsymbol{h}_{\ell-1}]\big)$$



4) CNN clásica (Conv + Pool + FC)

Visión · Convolución

$$Y[i,j] = \sum_{u,v} K[u,v]\, X[i+u, j+v], \qquad \text{Pool: } Y = \max_{\Omega} X$$



5) MobileNet (Depthwise Separable Conv)

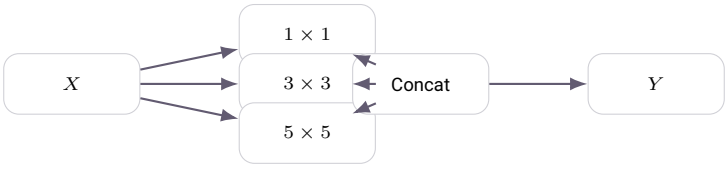
Eficiente · CNN

$$\text{DSConv}(X) = \underbrace{\text{DWConv}(X)}_{\text{por canal}} \rightarrow \underbrace{\text{PWConv}(1\times1)}_{\text{mezcla canales}}$$

6) Inception (ramas en paralelo)

CNN · Multi-escala

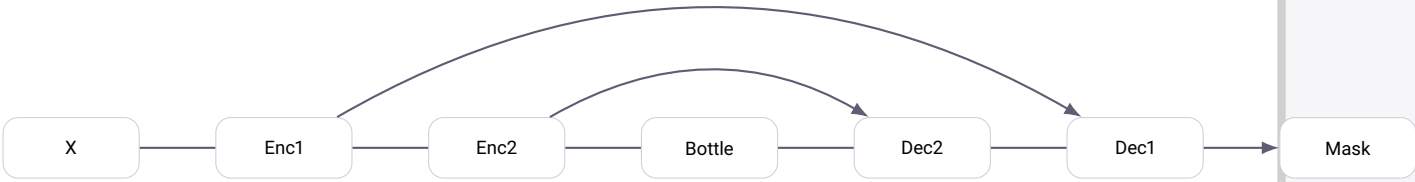
$$Y = \text{Concat}\left(f_{1\times1}(X), f_{3\times3}(X), f_{5\times5}(X), \text{pool}(X)\right)$$



7) U-Net (segmentación con skips)

Seg · Encoder-Decoder

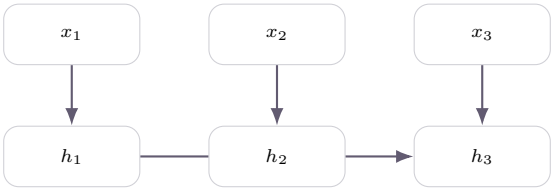
$$\text{Encoder}(X) \to \text{Bottleneck} \to \text{Decoder}(\cdot), \quad \text{skips: concat a misma escala}$$



8) RNN (recurrente)

Secuencias

$$\boldsymbol{h}_t = \tanh(\boldsymbol{W}_{xh}\boldsymbol{x}_t + \boldsymbol{W}_{hh}\boldsymbol{h}_{t-1} + \boldsymbol{b}), \quad \hat{\boldsymbol{y}}_t = g(\boldsymbol{h}_t)$$



9) GRU

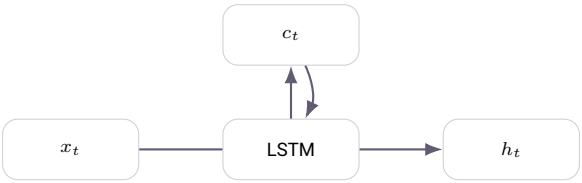
RNN · Gates

$$\boldsymbol{z}_t = \sigma(\boldsymbol{W}_z[\boldsymbol{h}_{t-1}, \boldsymbol{x}_t]), \; \boldsymbol{r}_t = \sigma(\boldsymbol{W}_r[\boldsymbol{h}_{t-1}, \boldsymbol{x}_t]), \quad \boldsymbol{h}_t = (1 - \boldsymbol{z}_t) \odot \boldsymbol{h}_{t-1} + \boldsymbol{z}_t \odot \tilde{\boldsymbol{h}}_t$$

10) LSTM

RNN · Memoria

$$\boldsymbol{c}_t = \boldsymbol{f}_t \odot \boldsymbol{c}_{t-1} + \boldsymbol{i}_t \odot \tilde{\boldsymbol{c}}_t, \quad \boldsymbol{h}_t = \boldsymbol{o}_t \odot \tanh(\boldsymbol{c}_t)$$



11) Seq2Seq (Encoder-Decoder)

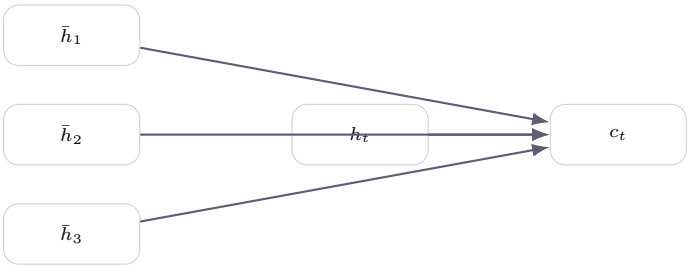
NLP · Traducción

Encoder: $\{\boldsymbol{x}_t\} \mapsto \boldsymbol{c}$, Decoder: $\boldsymbol{c} \mapsto \{\hat{\boldsymbol{y}}_t\}$

12) Atención (additive / Bahdanau)

Attention

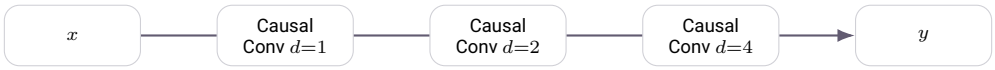
$$\alpha_{t,s} = \mathrm{softmax}_s(\text{score}(\boldsymbol{h}_t, \bar{\boldsymbol{h}}_s)), \quad \boldsymbol{c}_t = \sum_s \alpha_{t,s}\, \bar{\boldsymbol{h}}_s$$



13) TCN (Temporal Convolution Network)

Series · Causal

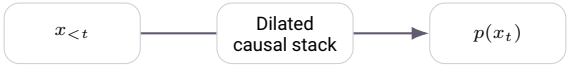
$$y_t = \sum_{i=0}^{k-1} w_i\, x_{t-d\cdot i} \quad \text{(conv causal con dilatación } d\text{)}$$



14) WaveNet (TCN generativa)

Audio · Autoregresiva

$$p(\boldsymbol{x}) = \prod_t p(x_t \mid x_{<t}), \quad \text{con convs causales dilatadas + gated activations}$$



15) ConvLSTM (espacio + tiempo)

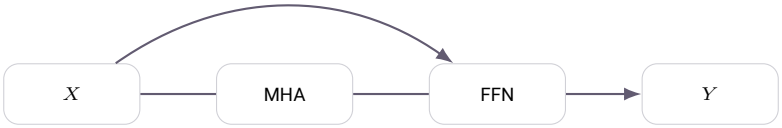
Video · RNN

LSTM donde $(\boldsymbol{W} * X)$ usa convoluciones ($*$) en lugar de matrices densas.

16) Transformer (self-attention)

Transformer · Atención

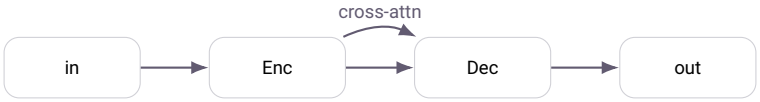
$$\mathrm{Attn}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}}\right)\boldsymbol{V}, \quad \boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}_Q,\ \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}_K,\ \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}_V$$



17) Variantes Transformer

Encoder · Decoder · Enc-Dec

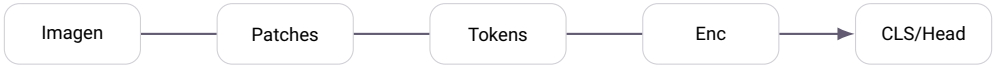
- **Encoder-only** (p.ej. BERT): representación para clasificación/retrieval.
- **Decoder-only** (p.ej. GPT): autoregresivo $p(x) = \prod_t p(x_t|x_{<t})$.
- **Encoder-Decoder** (p.ej. T5): seq2seq con cross-attention.



18) ViT (Vision Transformer)

Visión · Patches

$$X \in \mathbb{R}^{H \times W \times C} \rightarrow \text{patches} \rightarrow \text{tokens} \rightarrow \text{Transformer}$$



19) Swin (ventanas + jerarquía)

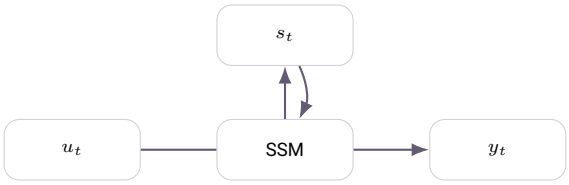
Visión · Windowed

Self-attn en ventanas locales $\Rightarrow$ costo menor + pirámide de escalas

20) SSM / Mamba-like (modelos de espacio de estados)

Largo contexto · SSM

$$\boldsymbol{s}_{t+1} = \boldsymbol{A}\boldsymbol{s}_t + \boldsymbol{B}\boldsymbol{u}_t, \qquad \boldsymbol{y}_t = \boldsymbol{C}\boldsymbol{s}_t + \boldsymbol{D}\boldsymbol{u}_t$$



21) Autoencoder (AE)

Compresión · Reconstrucción

$$\boldsymbol{z} = E_\theta(\boldsymbol{x}), \quad \hat{\boldsymbol{x}} = D_\phi(\boldsymbol{z}), \quad \mathcal{L} = \|\hat{\boldsymbol{x}} - \boldsymbol{x}\|_2^2$$

22) VAE (Variational Autoencoder)

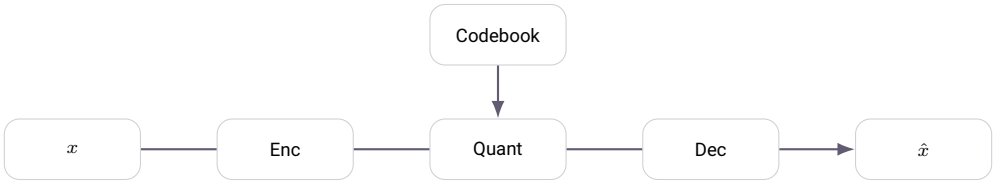
Generativo · Latente

$$\mathcal{L} = \mathbb{E}_{q_\theta(\boldsymbol{z}|\boldsymbol{x})}[-\log p_\phi(\boldsymbol{x}|\boldsymbol{z})] + \mathrm{KL}\left(q_\theta(\boldsymbol{z}|\boldsymbol{x}) \| p(\boldsymbol{z})\right)$$

23) VQ-VAE (latente discreto)

Codebook · Audio/Imagen

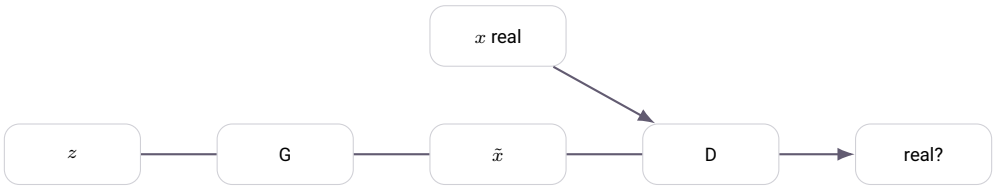
$$\boldsymbol{z}_e = E(\boldsymbol{x}), \quad \boldsymbol{z}_q = \mathrm{Quantize}(\boldsymbol{z}_e;\, \mathcal{C}), \quad \hat{\boldsymbol{x}} = D(\boldsymbol{z}_q)$$



24) GAN (Generative Adversarial Network)

Generativo · Adversarial

$$\min_G \max_D \; \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z}\sim p(\boldsymbol{z})}[\log(1 - D(G(\boldsymbol{z})))]$$



25) Diffusion / DDPM

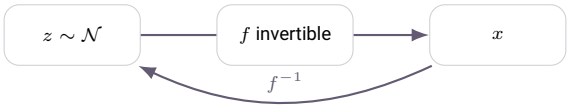
Generativo · Denoising

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1},\; \beta_t \boldsymbol{I}\Big), \quad p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \approx \text{denoise}_\theta(\boldsymbol{x}_t, t)$$

26) Normalizing Flows (invertibles)

Generativo · Exact LL

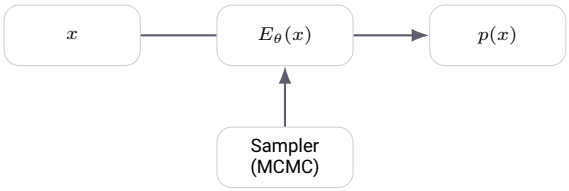
$$\log p(\boldsymbol{x}) = \log p(\boldsymbol{z}) + \log \left| \det \frac{\partial f^{-1}}{\partial \boldsymbol{x}} \right|, \quad \boldsymbol{z} = f^{-1}(\boldsymbol{x})$$



27) Energy-Based Models (EBM)

Energía · Sampler

$$p(\boldsymbol{x}) = \frac{e^{-E_\theta(\boldsymbol{x})}}{Z}, \qquad Z = \int e^{-E_\theta(\boldsymbol{x})}\, d\boldsymbol{x}$$



28) RBM / Boltzmann (clásico)

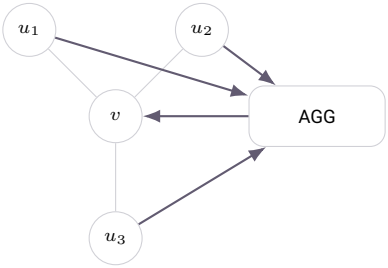
Energía · Latente

$$E(\boldsymbol{v},\boldsymbol{h}) = -\boldsymbol{b}^\top \boldsymbol{v} - \boldsymbol{c}^\top \boldsymbol{h} - \boldsymbol{v}^\top \boldsymbol{W} \boldsymbol{h}$$

29) GNN (GCN / GraphSAGE / MPNN)

Grafos · Message Passing

$$\boldsymbol{h}_v^{(k)} = \phi\Big(\boldsymbol{W}^{(k)}\ \mathrm{AGG}(\{\boldsymbol{h}_u^{(k-1)} : u \in \mathcal{N}(v) \cup \{v\}\})\Big)$$



30) GAT (Graph Attention Network)

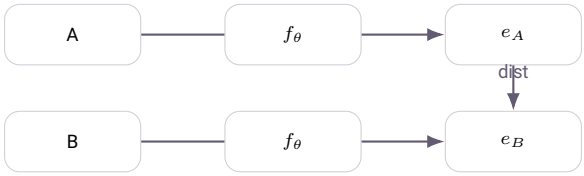
Grafos · Attention

$$\alpha_{vu} = \mathrm{softmax}_{u\in\mathcal{N}(v)}(a(\boldsymbol{W}\boldsymbol{h}_v, \boldsymbol{W}\boldsymbol{h}_u)), \quad \boldsymbol{h}'_v = \sum_{u\in\mathcal{N}(v)} \alpha_{vu}\, \boldsymbol{W}\boldsymbol{h}_u$$

31) Siamese / Metric Learning

Embeddings · Contraste

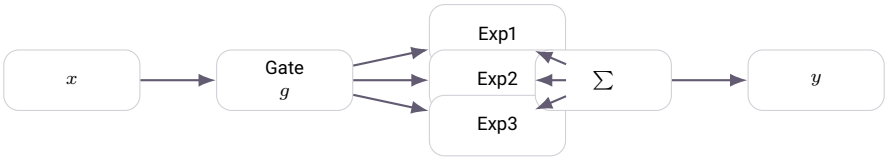
$$\mathcal{L}_{\text{triplet}} = \text{máx}\left(0,\ d(a,p) - d(a,n) + m\right)$$



32) Mixture of Experts (MoE)

Escalado · Ruteo

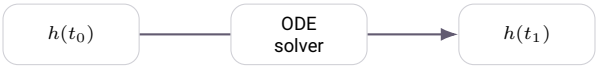
$$y = \sum_{i=1}^{M} g_i(\boldsymbol{x})\, f_i(\boldsymbol{x}), \quad \sum_i g_i(\boldsymbol{x}) = 1$$



33) Neural ODE

Continuo · Dinámica

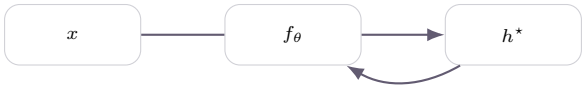
$$\frac{d\boldsymbol{h}(t)}{dt} = f_\theta(\boldsymbol{h}(t), t), \qquad \boldsymbol{h}(t_1) = \text{ODESolve}(\boldsymbol{h}(t_0), f_\theta)$$



34) Deep Equilibrium (DEQ)

Punto fijo · Memoria

$$\boldsymbol{h}^\star = f_\theta(\boldsymbol{h}^\star, \boldsymbol{x}) \quad \text{(se busca un punto fijo, no apilar capas)}$$



35) Spiking Neural Networks (LIF)

Neuromórfico · SNN

$$\tau\frac{dV}{dt} = -(V - V_{\text{rest}}) + RI(t), \quad \text{si } V \geq V_{\text{th}} \Rightarrow \text{spike y reset}$$